

Lecture 3: Matrix multiplication

Suppose B is an $n \times p$ matrix over a field F with rows $\beta_1, \dots, \beta_n$ and that from B we construct a matrix C with rows $\gamma_1, \dots, \gamma_m$ by forming certain linear combinations:

$$\gamma_i = A_{i1}\beta_1 + A_{i2}\beta_2 + \dots + A_{in}\beta_n \quad (1.4)$$

The rows of C are determined by the ~~mxn~~ $m \times n$ scalars A_{ij} which are themselves the entries of an $m \times n$ matrix A .
From (1.4),

$$(C_{i1} \quad \dots \quad C_{ip}) = \sum_{r=1}^n (A_{ir} \beta_{r1} \quad \dots \quad A_{ir} \beta_{rp}),$$

$$\text{entries of } C: C_{ij} = \sum_{r=1}^n A_{ir} B_{rj}.$$

Defⁿ: Let A be an $m \times n$ matrix over the field F and let B be an $n \times p$ matrix over F . The product AB is the $m \times p$ matrix C whose i, j entry is

$$C_{ij} = \sum_{r=1}^n A_{ir} B_{rj}.$$

Example: (a) Consider $\begin{bmatrix} 5 & -1 & 2 \\ 0 & 7 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 5 & -1 & 2 \\ 15 & 4 & 8 \end{bmatrix}$

$$\text{More, } \gamma_1 = (5 \quad -1 \quad 2) = 1 \cdot (5 \quad -1 \quad 2) + 0 \cdot (15 \quad 4 \quad 8)$$

$$\gamma_2 = (0 \quad 7 \quad 2) = -3 \cdot (5 \quad -1 \quad 2) + 1 \cdot (15 \quad 4 \quad 8).$$

$$(b) \begin{bmatrix} 0 & 6 & 1 \\ 9 & 12 & -8 \\ 12 & 62 & -3 \\ 3 & 8 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -2 & 3 \\ 5 & 4 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 6 & 1 \\ 3 & 8 & -2 \end{bmatrix}$$

$$\gamma_3 = 5 \cdot (0 \quad 6 \quad 1) + 4 \cdot (3 \quad 8 \quad -2)$$

B is an $n \times p$ matrix, $B = [B_1, \dots, B_p]$,
 $B_j = \begin{bmatrix} B_{1j} \\ \vdots \\ B_{nj} \end{bmatrix}$, $1 \leq j \leq p$. (B_j) is $1 \times n$ matrix.
 column matrix.

Check that $AB = [AB_1, \dots, AB_p]$.

Thm: If A, B, C are matrices over the field F such that the products BC and $A(BC)$ are defined, then so are the products AB , $(AB)C$, and
 $(AB)C = A(BC)$.

Proof: —————

Remark: For a square matrix A , A^n is well-defined
 $A^p A^q A^r = A^s A^t A^u$ for all $p+q+r=s+t+u=n$.

$A(BC) = (AB)C \rightarrow$ linear combinations of linear combinations of the rows of C are again linear combinations of the rows of C .

If $B \xrightarrow[\text{row operations}]{\text{elementary}} C$, then each row of C is a linear combination of the rows of B , and so $\exists$ a matrix A s.t. $AB = C$.
 (There can be many such A 's in general.)

Defⁿ: An $m \times m$ matrix is said to be an elementary matrix if it can be obtained from the $m \times m$ identity matrix $I_{m \times n}$ by means of a single elementary row operation.

Example: 2×2 elementary matrices:

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ c & 1 \end{bmatrix},$$

$$\begin{bmatrix} c & 0 \\ 0 & 1 \end{bmatrix} \text{ for } c \neq 0, \begin{bmatrix} 1 & 0 \\ 0 & c \end{bmatrix} \text{ for } c \neq 0.$$

Thm: Let e be an elementary row operation and let E be the $m \times n$ elementary matrix $E = e(A)$. Then, for every $m \times n$ matrix A , $e(A) = EA$.

Proof: Type (I) $E_{ik} = \begin{cases} \delta_{ik}, & i \neq r \\ \delta_{ik} + c\delta_{sk}, & i = r \end{cases}$ (To replace row r with row $s + c \cdot \text{row } s$)

$$(EA)_{ij} = \sum_{k=1}^m E_{ik} A_{kj} = \begin{cases} A_{ik}, & i \neq r \\ A_{rj} + cA_{sj}, & i = r. \end{cases}$$

Check for other types.

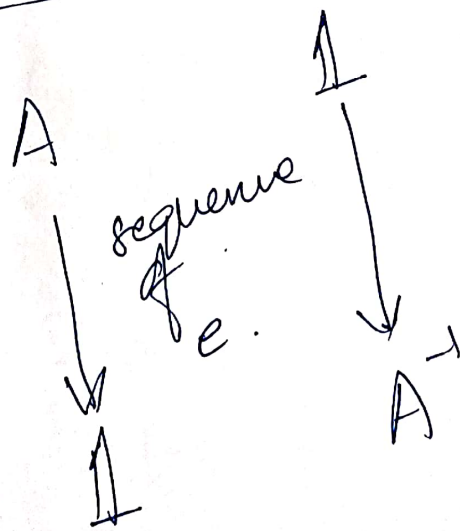
$E_{rr}, \begin{bmatrix} 1 & \dots & 1 & \dots & 1 \\ 0 & \dots & 0 & \dots & 0 \end{bmatrix}$ $E_{rs}, \begin{bmatrix} 1 & \dots & 1 & \dots & 1 \\ 0 & \dots & 0 & \dots & 0 \end{bmatrix}$ $E_{rs}, \begin{bmatrix} 1 & \dots & 1 & \dots & 1 \\ 0 & \dots & 0 & \dots & 0 \end{bmatrix}$

$r \rightarrow r + c \cdot s$

Corollary: A square matrix with either a left inverse or right inverse is invertible.

Proof: $A_{n \times n}$. Suppose left inverse of A exists, $BA = I$. Then, $AX = 0$ has only the trivial solⁿ, because $X = I X = B(AX)$. $\therefore A$ is invertible.

If A has a right inverse, $AC = I$. Then C has a left inverse & is therefore invertible. It then follows $A = C^{-1}$, so A is invertible w/ inverse C .



$$AX = Y$$

$$RX = Z,$$

$$\text{if } R = PA$$

$$\text{then } Z = PY$$

invertible matrix.

Corollary: let A and B be $n \times n$ matrices over the field F . Then B is row-equivalent to A if and only if $B = PA$, where P is a product of $n \times n$ elementary operations.

IV Invertible matrices.

Defn: let A be an $n \times n$ matrix over the field F . An $n \times n$ matrix B such that $BA = I$ is called a left inverse of A ; an $n \times n$ matrix B such that $AB = I$ is called a right inverse of A . If $AB = BA = I$ then B is called a two-sided inverse of A and A is said to be invertible.

Lemma: If A has a left inverse B and a right inverse C , then $B = C$.

Proof: $B = BI = BAC = IC = C$.

Thm: let A and B be $n \times n$ matrices over the field F .

(i) If A is invertible, so is A^T and $(A^{-1})^T = A^T$.

(ii) If both A and B are invertible, so is AB and $(AB)^{-1} = B^{-1}A^{-1}$.

Corollary: A product of invertible matrix is invertible.

Theorem: An elementary matrix is invertible.

Proof: _____

Thm. If A is an $n \times n$ matrix, the following are equivalent.

- (i) A is invertible.
- (ii) A is row-equivalent to $I_{n \times n}$.
- (iii) A is product of elementary operations.

Proof: _____

Thm: For an $n \times n$ matrix A , the following are equivalent.

- (i) A is invertible.
- (ii) The homogenous system $AX=0$ has only the trivial solⁿ.
- (iii) The system of eq's $AX=Y$ has a solⁿ X for each $n \times 1$ matrix Y .

Proof: _____

Column-equivalent

Column-reduced echelon matrix

~~Column~~ Elementary column operations: AE

$$\begin{aligned} AX &= Y \\ X^T A^T &= Y^T \end{aligned}$$